Summary in Graph

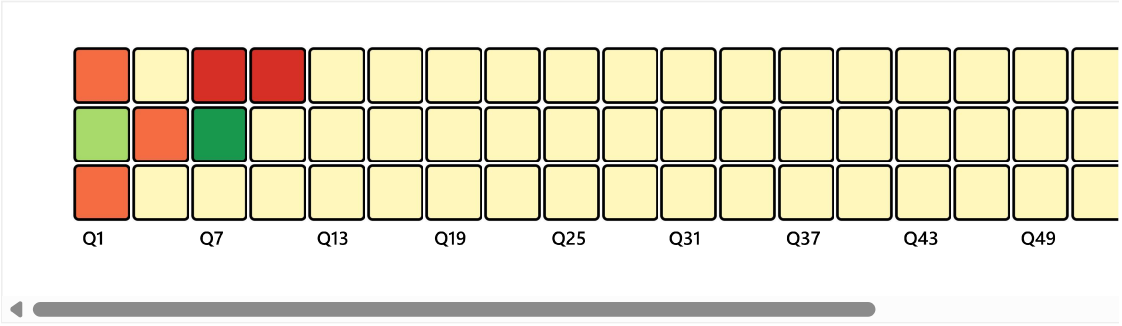
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 10)

Qs. Attempted:	**7** (4 + 3)	**Correct Marks:**	**3** (1 + 2)
Correct Attempts:	**2** (1 + 1)	**Penalty Marks:**	**2.33** (1 + 1.33)
Incorrect Attempts:	**5** (3 + 2)	**Resultant Marks:**	**0.67** (0 + 0.67)

Total Questions:	**65** (30 + 35)
Total Marks:	**100** (30 + 70)
Exam Duration:	**180 Minutes**
Time Taken:	**180 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (12) out of 120 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Members of the Rockham Soccer League buy socks and T-shirts. Socks cost $4 per pair and each T-shirt costs $5 more than a pair of socks. Each member needs one pair of socks and a shirt for home games and another pair of socks and a shirt for away games. If the total cost is $2366, how many members are in the League?

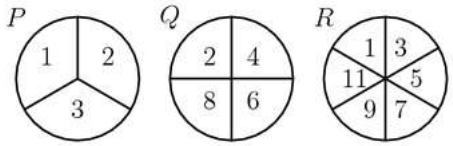
A. 77

B. 91
C. 143
D. 182

Your Answer: D | Correct Answer: B | Incorrect | Time taken: 01min 57sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Jeff rotates spinners P, Q and R and adds the resulting numbers. What is the probability that his sum is an odd number?



A. $\frac{1}{4}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. $\frac{2}{3}$

Your Answer: B | Correct Answer: B | Correct | Time taken: 02min 33sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Define the operation $x \heartsuit y$ as follows for all real numbers x and y:

$$x \heartsuit y = |x - y|$$

Which of the following statements is NOT true?

(A) $x \heartsuit y = y \heartsuit x$ for all x and y.

(B) $2(x \heartsuit y) = (2x) \heartsuit (2y)$ for all x and y.

(C) $x \heartsuit 0 = x$ for all x.

(D) $x \heartsuit x = 0$ for all x.

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 02min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

If $\log(xy^3) = 1$ and $\log(x^2y) = 1$ what is $\log(xy)$?

A. $-\frac{1}{2}$
B. 0
C. $\frac{1}{2}$
D. $\frac{3}{5}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 02min 45sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Fill in the blank with the correct idiom/phrase.

That boy from the town was a _____ in the sleepy village.

A. dog out of herd
B. sheep from the heap
C. fish out of water
D. bird from the flock

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 00min 16sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

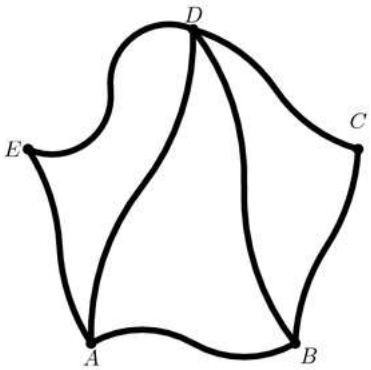
How many ordered pairs (a, b) such that a is a positive real number and b is an integer between 2 and 200 , inclusive, satisfy the equation $(\log_b a)^{2017} = \log_b(a^{2017})$?

A. 199
B. 398
C. 399
D. 597

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 02min 51sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Cities A, B, C, D, and E are connected by roads $\widetilde{AB}, \widetilde{AD}, \widetilde{AE}, \widetilde{BC}, \widetilde{BD}, \widetilde{CD}$, and $\widetilde{DE}$. How many different routes are there from A to B that use each road exactly once? (Such a route will necessarily visit some cities more than once.)



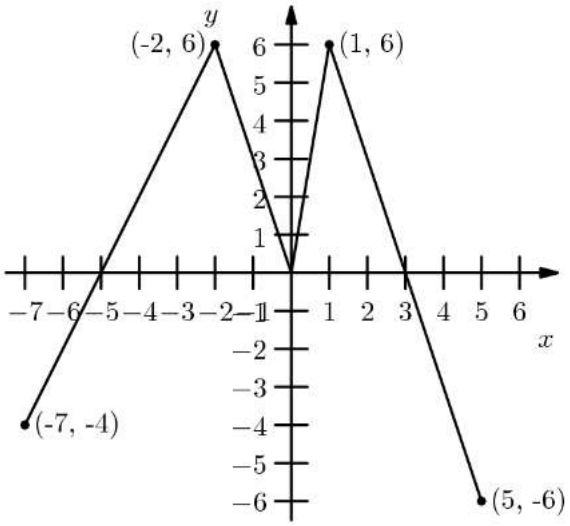
A. 7
B. 9

C. 12
D. 16

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 01min 41sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

The graph of the function f is shown below. How many solutions does the equation $f(f(x)) = 6$ have?



A. 2
B. 6
C. 5
D. 4

Your Answer: B | Correct Answer: B | Correct | Time taken: 02min 44sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Al, Bert, and Carl are the winners of a school drawing for a pile of Halloween candy, which they are to divide in a ratio of $3 : 2 : 1$, respectively. Due to some confusion they come at different times to claim their prizes, and each assumes he is the first to arrive. If each takes what he believes to be the correct share of candy, what fraction of the candy goes unclaimed?

A. $\frac{1}{18}$
B. $\frac{1}{6}$
C. $\frac{2}{9}$
D. $\frac{5}{18}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 13min 55sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

In the following sentence certain parts are underlined and marked P, Q, and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error.

The student corrected $\underbrace{\text{all the errors}}_{P}$ that $\underbrace{\text{the instructor marked}}_{Q}$ on $\underbrace{\text{the answer book}}_{R}$.

A. P
B. Q
C. R
D. No error

Your Answer: D | Correct Answer: B | Incorrect | Time taken: 02min 17sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

A. 0 and 0.5
B. 0 and 1
C. 0.5 and 1
D. 0.25 and 0.75

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider joint density $p\left(y_1, y_2\right)$

$y_2 \downarrow \ y_1 \rightarrow$	**1**	**2**	**3**
-1	0	0.4	0.1
-2	0.3	0.2	0

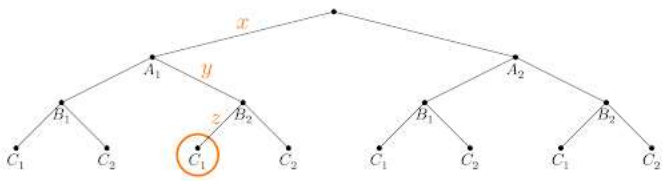
Let $F(y_1, y_2)$ be the cumulative distribution function (CDF).

What is $F(3, -1) =$

A. 0.2
B. 0.3
C. 0.4
D. 0.5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Others



Which of the following statement(s) is/are CORRECT regarding the probability tree diagram above?

A. The probability x represents $P(A_1)$.
B. The probability y represents $P(B_2)$.
C. The probability z represents $P(C_1 \mid B_2 \cap A_1)$.
D. The circled node represents the event $A_1 \cap B_2 \cap C_1$.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider the following joint pdf's for the random variables X and Y.

I. $f(x,y) = 4x^2y^3$
II. $f(x,y) = \frac{1}{2}\left(x^3y + xy^3\right)$
III. $f(x,y) = 6e^{-3x-2y}$

Which of the above distributions shows that X and Y are independent random variables?

A. Only I.
B. Only II and III.
C. Only I and III
D. All of I, II, and III.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider the following confusion matrix

		Predicted class		
		C_1	C_2	C_3
Actual Class	C_1	110	8	7
	C_2	16	130	10
	C_3	26	5	120

What is the recall for class C_1 ?

A. $110/125$
B. $110/152$
C. $110/172$
D. $110/432$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Assume that we have the following covariance matrix XX^T, please calculate the first principal component.

$$XX^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

A. $[1, 0, 0]^T$
B. $[0, 1, 0]^T$
C. $[0, 0, 1]^T$
D. $[1, 2, 3]^T$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following are valid expressions for the mean squared error objective function for linear regression with dataset $\mathcal{D} = \left\{\mathbf{x}^{(i)}, y^{(i)}\right\}_{i=1}^{N}$, with each $\mathbf{x}^{(i)} \in \mathbb{R}^M$ and the design matrix $\mathbf{X} \in \mathbb{R}^{N \times (M+1)}$. y and $\boldsymbol{\theta}$ are column vectors.
Select all that apply:

A. $J(\boldsymbol{\theta}) = \frac{1}{N}\|\mathbf{y} - \boldsymbol{\theta}\mathbf{X}\|_2^2$
B. $J(\boldsymbol{\theta}) = \frac{1}{N}\left\|\mathbf{y}^T - \boldsymbol{\theta}\mathbf{X}\right\|_2^2$
C. $J(\boldsymbol{\theta}) = \frac{1}{N}\left\|\mathbf{y}^T - \mathbf{X}\boldsymbol{\theta}\right\|_2^2$
D. $J(\boldsymbol{\theta}) = \frac{1}{N}\|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which two extreme points show the best performance and the worst performance respectively on the ROC curve?

A. $(1, 1), (0, 0)$
B. $(0, 1), (1, 0)$
C. $(1, 0), (0, 1)$
D. $(0, 1), (1, 1)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let $L(\boldsymbol{w}) = \frac{1}{2}\sum_i \left(y_i - \boldsymbol{w}^\top \boldsymbol{\phi}\left(x_i\right)\right)^2$ be the loss function corresponding to a linear regression problem. Which equation represents the stochastic gradient descent update for $\boldsymbol{w}$?

A. $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} - \eta\left(y_i - \boldsymbol{w}^\top \boldsymbol{\phi}\left(\boldsymbol{x}_i\right)\right)\boldsymbol{\phi}\left(x_i\right)$
B. $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} - \eta\sum_i\left(y_i - \boldsymbol{w}^\top \boldsymbol{\phi}\left(x_i\right)\right)\boldsymbol{\phi}\left(x_i\right)$
C. $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} - \eta\left(y_i - \operatorname{sign}\left(\boldsymbol{w}^\top \boldsymbol{\phi}\left(x_i\right)\right)\right)\boldsymbol{\phi}\left(x_i\right)$, where $\operatorname{sign}(\cdot)$ is the sign function
D. $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} - \eta\left(y_i - \sigma\left(\boldsymbol{w}^\top \boldsymbol{\phi}\left(x_i\right)\right)\right)\boldsymbol{\phi}\left(x_i\right)$, where $\sigma(z) = 1/\left(1 + e^{-z}\right)$ is the sigmoid function.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Find $T\begin{bmatrix}1\\1\end{bmatrix}$ if $T:\mathbb{R}^2\to\mathbb{R}^3$ is a linear transformation such that

$$T\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}2\\0\\6\end{bmatrix}\quad\text{and}\quad T\begin{bmatrix}3\\4\end{bmatrix}=\begin{bmatrix}0\\-2\\4\end{bmatrix}$$

A. $\begin{bmatrix}2\\2\\2\end{bmatrix}$

B. $\begin{bmatrix}2\\-2\\10\end{bmatrix}$

C. $\begin{bmatrix}1\\-1\\5\end{bmatrix}$

D. $\begin{bmatrix}-1\\-1\\-1\end{bmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider the linear system

$$\begin{cases}x_1+x_2=3\\x_1-x_2=2\end{cases}$$

According to Cramer's rule, what is x_2 ?

A. $\dfrac{\begin{vmatrix}1&3\\1&2\end{vmatrix}}{\begin{vmatrix}1&1\\1&-1\end{vmatrix}}$

B. $\begin{vmatrix}1&1\\1&-1\end{vmatrix}$

C. $\dfrac{\begin{vmatrix}1&0\\1&1\end{vmatrix}}{\begin{vmatrix}1&1\\1&-1\end{vmatrix}}$

D. $\dfrac{\begin{vmatrix}1&1\\1&-1\end{vmatrix}}{\begin{vmatrix}1&1\\1&-1\end{vmatrix}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose that the null space of a matrix A has dimension 3. Which of the following must be true about A ?

A. $A\vec{x}=\vec{b}$ always has a solution for any $\vec{b}$.
B. $A\vec{x}=\vec{b}$ never has a solution for any $\vec{b}$.
C. If $A\vec{x}=\vec{b}$ has a solution, it will be unique.
D. If $A\vec{x}=\vec{b}$ has a solution, it will not be unique.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose that only numbers in $1 \ldots 100$ appear as keys in a binary search tree. While searching for 50, which of the following sequences of keys could not be examined?

A. $10, 30, 70, 60, 50$
B. $100, 20, 80, 30, 50$
C. $1, 100, 20, 70, 50$
D. $10, 40, 70, 30, 50$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

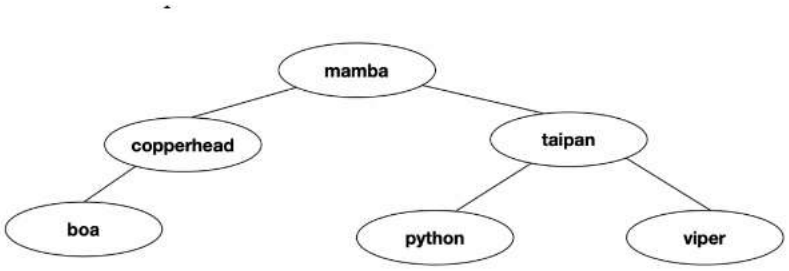
Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following correctly shows the iterations of an ascending (from left to right) insertion sort on an array with the following elements: $\{7, 3, 8, 5, 2\}$?

A. $\{2, 3, 8, 5, 7\}, \{2, 3, 8, 5, 7\}, \{2, 3, 5, 8, 7\}, \{2, 3, 5, 7, 8\}$
B. $\{3, 7, 8, 5, 2\}, \{3, 7, 8, 5, 2\}, \{3, 5, 7, 8, 2\}, \{2, 3, 5, 7, 8\}$
C. $\{3, 7, 8, 5, 2\}, \{3, 5, 7, 8, 2\}, \{2, 3, 5, 7, 8\}$
D. $\{2, 3, 8, 5, 7\}, \{2, 3, 5, 8, 7\}, \{2, 3, 5, 7, 8\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others



Consider these two traversals of the tree shown:

1. boa, copperhead, python, viper, taipan, mamba

2. mamba, copperhead, boa, taipan, python, viper

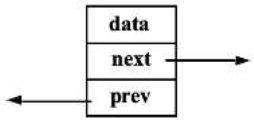
Which one statement below is true about these traversals?

A. 1 is postorder, 2 is preorder
B. 1 is postorder, and 2 is not preorder
C. 1 is not postorder, and 2 is preorder
D. 1 is not postorder, and 2 is not preorder

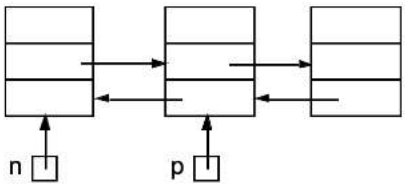
Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Nodes for a doubly linked list are defined to have the following structure:



The next instance variable stores a reference to the next node in the list, and the prev instance variable refers to the previous node in the list.
Below is a list of three of these nodes, along with two reference variables, n and p, that refer to specific nodes in the list.



Which of the following expressions does not refer to the third node in the list?

A. n.next.next
B. p.prev.next
C. p.next.prev.next
D. n.next.next.prev.next

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Others

Consider the below Python function:

```
def mystery(l):
    return [] if len(l) < 2 else mystery(l[1:])
```

Given the following function call:

```
mystery(['a','b','c','d','e','f','g','h'])
```

What is the length of the output list returned by the above function call?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Given the array of integers arr shown below:

```
arr =[5,19,6,12,48,6,23,39,2,18]
```

What is the output of the following Python code?

```
    a2 = arr.copy()
    a2[3] = arr[2]
    arr[2] = arr[3]
    arr[3] = a2[3]
5. print(arr[2],arr[3])
```

A. 6 12
B. 12 6
C. 6 6
D. None of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Evaluate $\lim_{x\to 2} \dfrac{1+\sqrt{x+2}}{1-\sqrt{2x}}$

A. $\frac{-3}{4}$
B. -3
C. -5
D. $1+\sqrt{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

The function

$$G(x) = \begin{cases} x-5 & x<2 \\ -5 & x=2 \\ 5x-13 & x>2 \end{cases}$$

is NOT continuous at $x=2$ because

A. $G(2)$ is not defined
B. $\lim_{x\to 2} G(x)$ does not exist
C. $\lim_{x\to 2} \neq G(2)$
D. $G(2) \neq -5$

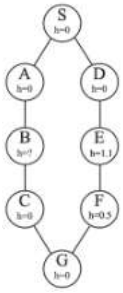
Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose that you are designing a heuristic h for the following graph. Assume all edges in the graph have cost 1. You are told that

- $h(F) = 0.5$
- $h(E) = 1.1$
- All other heuristic values except $h(B)$ are fixed to zero.

What is the range of values for $h(B)$ that yield an admissible heuristic AND result in the given expansion ordering "B expanded before E expanded before F" when using A graph search? Break ties alphabetically.

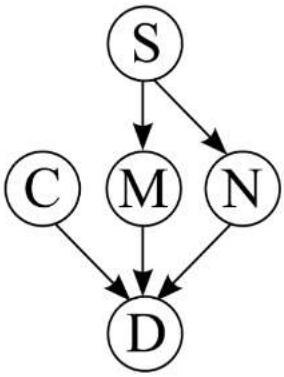


A. $0.0 \leq h(B) \leq 1.1$
B. $1.1 < h(B) \leq 1.5$
C. $-1.1 < h(B) \leq 1.1$
D. The given ordering is impossible with an admissible heuristic.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 1 | Penalty: 0 | Others

Given the following Bayesian Network, consider that prior sampling is used to draw the samples below:



S	C	M	N	D
$-s$	$+c$	$+m$	$-n$	$+d$
$+s$	$+c$	$-m$	$-n$	$-d$
$+s$	$-c$	$-m$	$+n$	$-d$
$+s$	$+c$	$+m$	$-n$	$+d$
$+s$	$+c$	$-m$	$+n$	$+d$
$+s$	$-c$	$-m$	$+n$	$-d$
$+s$	$-c$	$-m$	$-n$	$-d$
$+s$	$+c$	$+m$	$+n$	$+d$
$+s$	$+c$	$-m$	$+n$	$-d$
$-s$	$-c$	$-m$	$-n$	$-d$

Based on rejection sampling using the samples above, what is $\mathrm{P}(-\mathrm{m} \mid +\mathrm{d}, +\mathrm{s})$? (rounded off upto 2 decimal places)

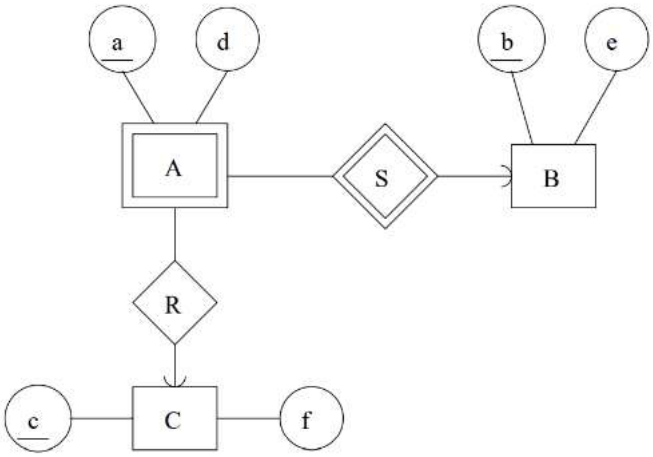
Your Answer: | Correct Answer: 0.33;0.34 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Others

In the relation $R(A, B, C, D, E)$ with functional dependencies $AB \rightarrow C, C \rightarrow B$, and $D \rightarrow E$, the number of superkeys is:

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others



Consider the above image (ER diagram) and it's convertion to a relational database schema, for the below question.
Which of the following entity sets or relationships does not get converted to a relation?

A. B
B. C
C. R
D. S

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Others

Which of the following statements is/are true for search algorithms?

A. If all moves are unit cost, breadth-first search always finds the optimal path to the goal.
B. Iterative-deepening search always finds the optimal path to the goal.
C. In a graph with no loops, depth-first search always finds the goal quicker than breadth-first search.
D. All admissible heuristics are equal. That is, A^* will search states in the same order for all admissible heuristics.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Inorder traversing a tree resulted $\mathrm{E\ A\ C\ K\ F\ H\ D\ B\ G}$; the preorder traversal would return.

A. $\mathrm{F\ A\ E\ K\ C\ D\ B\ H\ G}$
B. $\mathrm{F\ A\ E\ K\ C\ D\ H\ G\ B}$
C. $\mathrm{E\ A\ F\ K\ H\ D\ C\ B\ G}$
D. $\mathrm{F\ E\ A\ K\ D\ C\ H\ B\ G}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Which statement is NOT true about the 95% confidence level?

A. Confidence intervals computed by using the same procedure will include the true population value for 95% of all possible random samples taken from the population.
B. The procedure that is used to determine the confidence interval will provide an interval that includes the population parameter with probability of 0.95 .
C. The probability that the true value of the population parameter falls between the bounds of an already computed confidence interval is roughly 95%.
D. If we consider all possible randomly selected samples of the same size from a population, the 95% is the percentage of those samples for which the confidence interval includes the population parameter.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let X and Y be two random variables with the joint probability density function:

$$f(x,y) = \begin{cases} 2 & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

What is the conditional probability $P\left(X \leqslant \frac{2}{3} \middle| Y = \frac{3}{4}\right)$?

A. $\frac{5}{9}$
B. $\frac{2}{3}$
C. $\frac{7}{9}$
D. $\frac{8}{9}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Let X be a uniform random variable on the interval $(0, 10)$.
Find the conditional means $E(X \mid X \leq 6)$ and $E(X \mid X > 8)$, and the conditional variances $\mathrm{Var}(X \mid X \leq 6)$ and $\mathrm{Var}(X \mid X > 8)$.

A. $E(X \mid X \leq 6) = 3$
B. $\mathrm{Var}(X \mid X \leq 6) = 3$
C. $E(X \mid X > 8) = 9$
D. $\mathrm{Var}(X \mid X > 8) = 6$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the following sample space and random variables:

$$S = \{\omega_1, \omega_2, \omega_3\}, \quad P(\omega_i) = \frac{1}{3}, \quad \text{for } i = 1, 2, 3$$

Let the random variables X, Y, and Z be defined as:

$$X(\omega_1)=1, \quad X(\omega_2)=2, \quad X(\omega_3)=3$$
$$Y(\omega_1)=2, \quad Y(\omega_2)=3, \quad Y(\omega_3)=1$$
$$Z(\omega_1)=3, \quad Z(\omega_2)=1, \quad Z(\omega_3)=2$$

Find the probability mass function (PMF) of $M=X+Y-Z$.

A. $P(M=0)=\frac{1}{3}, P(M=2)=\frac{1}{3}, P(M=4)=\frac{1}{3}$
B. $P(M=0)=\frac{1}{4}, P(M=2)=\frac{1}{4}, P(M=4)=\frac{1}{2}$
C. $P(M=0)=\frac{1}{3}, P(M=2)=\frac{1}{2}, P(M=4)=\frac{1}{6}$
D. $P(M=0)=\frac{1}{2}, P(M=2)=\frac{1}{4}, P(M=4)=\frac{1}{4}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the bivariate data $(4,10),(-1,3),(0,2)$ which is assumed to arise from the model $y_i = b\,|x_i-3|+e_i$, where b is a constant and e_i is the noise of the i-th point, with e_i being independent and having mean 0 .
Given the above data, determine the least squares estimate for b.

A. $\frac{14}{13}$
B. 56
C. $\frac{13}{14}$
D. $\frac{13}{52}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the following training data set for a classification problem:

x	y
-1	-1
0	+1
1	+1
2	-1

The data is linearly separable in the new feature space $\phi(x)=\left[x, x^2,(x-1)^2\right]^\top$. Suppose we use a linear classifier with $\theta=[-1,-1,-1]^\top$ and an unknown intercept θ_0. To classify the data correctly, the condition $\theta^\top\phi(x)+\theta_0>0$ for the data points labeled +1 and $\theta^\top\phi(x)+\theta_0<0$ for the data points labeled -1 must be satisfied.
Which of the following values for θ_0 results in a correct separator for the training data?

A. $\theta_0=1$
B. $\theta_0=3$
C. $\theta_0=5$
D. $\theta_0=-1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Numerical Type | Award: 2 | Penalty: 0 | Others

Suppose we fit decision trees of varying depths to predict y using x_1 and x_2. The entire training set is shown in the table below. What is the lowest possible entropy of a node in a fitted tree with depth 1 (two leaf nodes)?

x_1	x_2	y
A	1	0
A	2	1
B	3	0
B	4	1
A	1	0
A	2	1
B	3	0
B	4	1

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Select the true statements about the decision boundaries obtained by linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA).

A. If we add extra features, LDA could produce a decision boundary that is nonlinear relative to the original input features.
B. QDA never produces a linear decision boundary.
C. Sometimes the QDA decision boundary is an ellipsoid.
D. Consider using LDA for a two-class classification problem in which both classes have prior probability 0.5 . LDA computes estimated means $\hat{\mu}_{\mathrm{C}}$ and $\hat{\mu}_{\mathrm{D}}$ for the two classes; suppose they are not equal. The midpoint halfway between $\hat{\mu}_{\mathrm{C}}$ and $\hat{\mu}_{\mathrm{D}}$ lies on the decision boundary obtained from LDA.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Let $S = \left\{\left(\mathbf{x}^{(1)}, y^{(1)}\right), \cdots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right\}$ be n linearly separable points by a separator through the origin in $\mathbb{R}^d$. Let S' be generated from S as:

$$S' = \left\{\left(c\mathbf{x}^{(1)}, y^{(1)}\right), \cdots, \left(c\mathbf{x}^{(n)}, y^{(n)}\right)\right\}$$

where $c > 1$ is a constant. Suppose that we would like to run the perceptron algorithm on both data sets separately, and that the perceptron algorithm converges on S. The initial weight is zero for both perceptrons. Which of the following statements are true?

A. The mistake bound of perceptron on S' is larger than the mistake bound on S
B. The perceptron algorithm when run on S and S' returns the same classifier, modulo constant factors (i.e., if $\mathbf{w}_S$ and $\mathbf{w}_{S'}$ are outputs of the perceptron for S and S', then $\mathbf{w}_S = c_1\mathbf{w}'_S$ for some constant c_1)
C. The perceptron algorithm converges on S'
D. The mistake bound remains unchanged

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the lined L spanned by the vector $\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. The distance from the vector $\vec{x} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$ to the line L is

A. $\sqrt{45}$
B. $\sqrt{5}$
C. $2\sqrt{3}$
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let $W = \text{Span}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$, where

$$\vec{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{w}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{w}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Apply the Gram-Schmidt orthogonalization process to construct an orthogonal basis $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ for W, where:
1. $\vec{v}_1$ is the first vector in the orthogonal basis, obtained by setting $\vec{v}_1 = \vec{w}_1$.
2. $\vec{v}_2$ is the second vector in the orthogonal basis, obtained by orthogonalizing $\vec{w}_2$ against $\vec{v}_1$.
3. $\vec{v}_3$ is the third vector, obtained by orthogonalizing $\vec{w}_3$ against both $\vec{v}_1$ and $\vec{v}_2$.

What is $\vec{v}_2$?

A. $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$
B. $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$
C. $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
D. $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -2 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let A be an $m \times n$ matrix with linearly independent columns and let $\vec{b}$ in $\mathbb{R}^m$ be a vector which is not in $\text{Col}(A)$. Which of the following statements may be false?

A. There exists a vector $\vec{x}$ in $\mathbb{R}^n$ with $A\vec{x} - \vec{b}$ perpendicular to $\text{Col}(A)$.
B. $\det(A^T A) \neq 0$.
C. The vector $\vec{b}$ is not the zero vector.

D. $\det(AA^T) \neq 0$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Which of the following statements are always true for an $n \times n$ matrix A ?

I. If A is invertible, then 0 is not an eigenvalue of A.
II. If $\mathbb{R}^n$ has a basis of eigenvectors of A, then A is diagonalizable.
III. Every matrix similar to A has the same characteristic polynomial as A.

A. I, II only
B. I, III only
C. II, III only
D. I, II, III

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Numerical Type | Award: 2 | Penalty: 0 | Others

Suppose A is a 3×4 matrix and B is a 4×5 matrix such that $AB = 0$. Given that the null space of A, denoted $\boldsymbol{N}(A)$, contains the column space of B, denoted $C(B)$, what is the maximum value of $\text{rank}(A) + \text{rank}(B)$?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider a square invertible matrix A. Let $\sigma_{\max}(A)$ be the largest singular value of A. What is the relationship between $\sigma_{\max}\left(A^{-1}\right)$ and $\sigma_{\max}(A)$?

A. $\sigma_{\max}\left(A^{-1}\right)\sigma_{\max}(A) \geq 1$
B. $\sigma_{\max}\left(A^{-1}\right)\sigma_{\max}(A) \leq 1$
C. $\sigma_{\max}\left(A^{-1}\right)\sigma_{\max}(A) = 1$
D. $\sigma_{\max}\left(A^{-1}\right)\sigma_{\max}(A) \geq \sigma_{\max}(A)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Suppose we create a variant of Dijkstra's algorithm called relaxRandomlyThenDijkstras. This variant works by first relaxing $E/2$ edges in a random order, then calls Dijkstra's as normal. If an edge is relaxed from a vertex whose current distance is infinite, the relaxation operations have no effect.
Which of the following can we say about this new algorithm? Assume there are no negative edges.

A. relaxRandomlyThenDijkstras always correctly computes the shortest paths tree (SPT) from s.
B. relaxRandomlyThenDijkstras correctly computes the shortest paths tree from s so long as none of the randomly relaxed edges are part of the SPT.
C. relaxRandomlyThenDijkstras correctly computes the shortest paths tree from s so long as all of the randomly relaxed edges are part of the SPT.

D. relaxRandomlyThenDijkstras never correctly computes the shortest paths tree (SPT) from s.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider two statements $\mathrm{S1}$ and $\mathrm{S2}$ below.

- $\mathrm{S1}$: If $f_1(n) = O(g_1(n))$ and $f_2(n) = O(g_2(n))$, then $f_1(n) + f_2(n) = O(\max\{g_1(n), g_2(n)\})$.
- $\mathrm{S2}$: If $f_1(n) = \Omega(g_1(n))$ and $f_2(n) = \Omega(g_2(n))$, then $f_1(n) + f_2(n) = \Omega(\min\{g_1(n), g_2(n)\})$.

Which of the following is correct?

A. $\mathrm{S1}$ is true but $\mathrm{S2}$ is false
B. $\mathrm{S1}$ is false but $\mathrm{S2}$ is true
C. Both statements are true
D. Both statements are false

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Professor Takis claims that he has designed an implementation of a special stack S that supports the following operations.

- $\mathrm{Push}(\mathrm{S}, x)$: usual push
- $\mathrm{Pop}(\mathrm{S}, x)$: usual pop
- $\mathrm{Delete\,Min}(\mathrm{S})$: remove from S the minimum element

He further claims that all the operations take $O(1)$ time.
Which of the following is correct about the above implementation?

A. Professor Takis's claim is correct if he has used linked list to implement given special stack
B. Professor Takis's claim is correct if he has used one array to implement given special stack
C. Professor Takis's claim is correct if he has used two arrays to implement given special stack
D. Professor Takis's claim can never be correct irrespective of data structures used to implement special stack

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the following randomized algorithm (pseudocode), which prints out some number of asterisks (*):

```
def randomizedSmile(A):
    n = len(A)
    if n <= 1:
        return

    for i in range(n):
        print("*")

    # Choose a uniformly random integer p in {1, ..., n-1}

    randomizedSmile(A[:p])
    randomizedSmile(A[p:])
```

What is the expected number of *'s that `randomizedSmile` prints out?

A. $O(1)$
B. $O(\log n)$
C. $O(n)$
D. $O(n \log n)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Consider two programs given below. Consider a non-empty linked list, which is either a circular or singly linked list. P1 and P2 are two programs that try to check if the linked list is circular or not. Let head point to the first node of the linked list.
- P1 takes the head of the linked list as its argument.
- P2 takes the head as the first argument and head. next as the second argument at the first call.

```
P1:
def P1(head):
    if head is none:
        return 1
    next_node = head.next
    while next_node is not None and next_node != head:
        next_node = next_node.next
    return 1 if next_node == head else 0
```

```
P2:
def P2(head, cur):
    if head is None:
        return 1
    if cur is None:
        return 0
    if head==cur:
        return 1
    return P2(head, cur.next)
```

Which of the following is/are true?

A. P1 returns 1 if the given linked list is circular.
B. P2 returns 1 if the given linked list is circular.
C. P1 returns 1 if the given linked list has an even number of nodes.
D. P2 returns 1 if the given linked list has an even number of nodes.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Assume f is an odd, continuous function with these values:

x	1	2	3
$f(x)$	-3	3	1

On which of the following intervals does the equation

$$f(x) = 0$$

have at least one solution?
More than one of these intervals may be correct.

A. $-3 \leq x \leq -2$
B. $-2 \leq x \leq -1$
C. $-1 \leq x \leq 1$
D. $1 \leq x \leq 2$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider the function

$$f(x_1, x_2, x_3) = x_1^2 + 6x_1x_2 + 3x_2^2 + 2x_3^2$$

Which statement is true?

A. f is not a quadratic form
B. f is a positive definite quadratic form
C. f is an indefinite quadratic form
D. f is a negative definite quadratic form

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider a state space where the initial state is the number 1 and each state k has two successors: numbers $2k$ and $2k + 1$. Suppose the goal state is 2010 . What is the branching factor in each direction of the bidirectional search?

A. forward: 1 , backward: 1
B. forward : 1, backward : 2
C. forward: 2, backward : 1
D. forward : 2, backward : 2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

In the following relational algebra expressions, R and S have the same schema, which includes attribute a, but the schemas are otherwise arbitrary.

$$Q_1: \quad \pi_a(R) - \pi_a(S)$$
$$Q_2: \quad \pi_a(R - S)$$

A. Q_1 and Q_2 produce the same answer.
B. The answer to Q_1 is always contained in the answer to Q_2.
C. The answer to Q_2 is always contained in the answer to Q_1.
D. Q_1 and Q_2 produce different answers.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

We have relations $R(a, b)$ and $S(a, c)$.
Suppose the following SQL statement holds:

```
NOT EXISTS (SELECT * FROM R, S WHERE R.a = S.a AND R.b<>S.c)
```

Which of the following combinations of tuples are prohibited from existing at the same time?

A. $(1, 1)$ in R and $(2, 2)$ in S.
B. $(1, 2)$ in R and $(2, 1)$ in S.
C. $(1, 2)$ in R and $(1, 2)$ in S.
D. $(1, 1)$ in R and $(1, 2)$ in S.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let $p(x), q(x)$, and $r(x)$ denote the following open statements.

$$\begin{aligned} p(x): &\quad x^2 - 8x + 15 = 0 \\ q(x): &\quad x \text{ is odd} \\ r(x): &\quad x > 0 \end{aligned}$$

For the universe of all integers, which of the following statements is/are false?

A. $\forall x[p(x) \rightarrow q(x)]$
B. $\forall x[q(x) \rightarrow p(x)]$
C. $\exists x[p(x) \rightarrow q(x)]$
D. $\exists x[q(x) \rightarrow p(x)]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

For a given predicate $\mathrm{P(x)}$, you might believe that the statements $\forall x\mathrm{P(x)}$ or $\exists \mathrm{xP(x)}$ are either true or false. $\exists x(\neg P(x))$ is false if and only if

A. $\exists \mathrm{x(P(x))}$ is true.
B. $\forall \mathrm{x(P(x))}$ is true.
C. $\forall \mathrm{x}(\neg \mathrm{P(x)})$ is true.
D. $\forall x(\neg P(x))$ is false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A simple random sample of size 10 from $N\left(\mu, \sigma^2\right)$ gives 98% confidence interval (20.49, 23.51).
Then the null hypothesis $H_0 : \mu = 20.5$ against $H_A : \mu \neq 20.5$

A. can be rejected at 2% level of significance
B. cannot be rejected at 5% level of significance
C. can be rejected at 10% level of significance
D. cannot be rejected at any level of significance

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let X be a random variable whose distribution depends on two conditions: C_1 and C_2, where $C_1 = C_2^c$.
Given that $P\left(C_1\right) = 1/4$ and the conditional CDFs are

$$F_X\left(x \mid C_1\right) = \begin{cases} 0 & \text{for } x < 0 \\ x/2 & \text{for } 0 \leq x \leq 2 \\ 1 & \text{for } x > 2 \end{cases}$$

and

$$F_X\left(x \mid C_2\right) = \begin{cases} 0 & \text{for } x < 0 \\ x/3 & \text{for } 0 \leq x \leq 3 \\ 1 & \text{for } x > 3 \end{cases}$$

Find the probability $P(X \leq 2)$.

A. $1/4$
B. $3/4$
C. $1/2$
D. $1/3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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